Project Report: Credit Card Fraud Detection & Risk Analytics

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Tools Used: Microsoft Excel, Power BI, DAX

Domain: Financial Services & Risk Management

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1. Executive Summary

Credit card fraud is a critical challenge for financial institutions, leading to direct monetary losses and compromised customer trust. This project outlines the development of a diagnostic and descriptive analytics dashboard using Microsoft Power BI. By ingesting and modeling transactional data, the resulting dashboard empowers risk management teams to visualize geographic fraud hotspots, identify compromised merchants, and analyze suspicious transaction behaviors (such as high-value purchases or anomalous refunds) to accelerate fraud mitigation.

2. Introduction & Problem Statement

2.1 The Problem

Traditional fraud detection relies heavily on automated algorithms that block transactions in real-time. However, risk analysts still require post-incident visibility to understand *how* and *where* the algorithms are failing, or to manually investigate suspicious batches of data. Reviewing raw logs of `TransactionID`s and `Amount`s in spreadsheets is highly inefficient and prone to human error.

2.2 Project Objective

The objective of this project is to transform raw transactional data into an interactive, visual risk-monitoring tool. The dashboard allows analysts to rapidly filter out legitimate transactions and focus entirely on anomalous patterns across different cities, transaction types, and merchants.

3. Data Profile & Architecture

The project utilizes a transactional dataset containing the following key attributes:

- **Unique Identifiers:** `TransactionID`, `MerchantID`.
- **Temporal Data:** `TransactionDate` (Timestamp of the event).
- **Financial Data:** `Amount` (The monetary value of the transaction).
- **Categorical Dimensions:** * `TransactionType` (Purchase, Refund).
    - `Location` (Geographic origin, e.g., New York, Dallas, San Antonio).
- **Target Variable:** `IsFraud` (Binary indicator where 1 = Fraudulent, 0 = Legitimate).

4. Methodology

The project followed a standard ETL (Extract, Transform, Load) and data modeling pipeline:

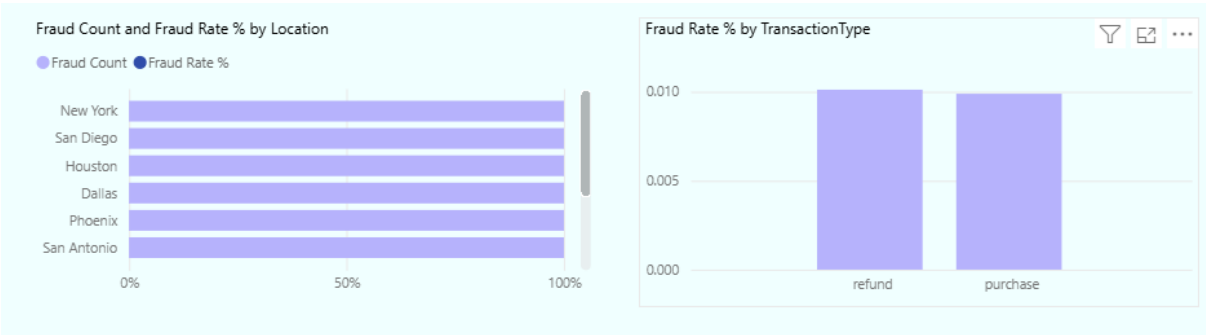
Phase 1: Data Preprocessing (Excel & Power Query)

- **Data Cleansing:** Checked for missing values within the `Amount` and `MerchantID` columns to ensure calculation accuracy.
- **Temporal Transformation:** The `TransactionDate` column contained combined date/time values (e.g., `15:35.5`). This was parsed and split in Power Query to allow for time-series analysis (e.g., identifying if fraud spikes during specific hours of the day).
- **Data Typing:** Ensured the `IsFraud` flag was formatted as a whole number (for mathematical aggregation) and the `Amount` as a fixed decimal/currency.

Phase 2: Data Modeling & DAX (Power BI)

Custom DAX (Data Analysis Expressions) measures were authored to create dynamic risk KPIs that recalculate instantly when analysts apply filters.

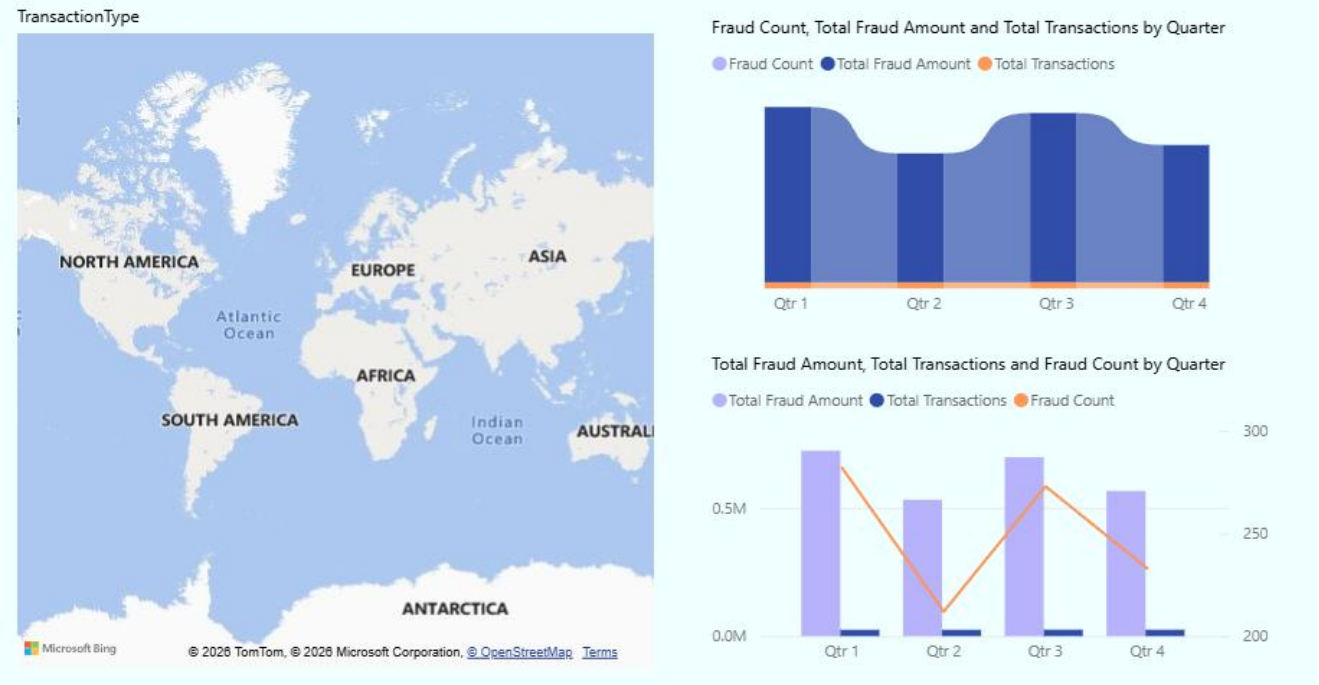
- **Total Transactions:** `COUNTROWS('FraudData')`
- **Fraud Volume:** `CALCULATE(COUNTROWS('FraudData'), 'FraudData'[IsFraud] = 1)`
- **Financial Exposure (Total Fraud Amount):** `CALCULATE(SUM('FraudData'[Amount]), 'FraudData'[IsFraud] = 1)`
- **Fraud Rate (%):** `DIVIDE([Fraud Volume], [Total Transactions], 0)`



Phase 3: Dashboard Design & Visualization

The UI/UX was designed specifically for risk analysts, utilizing a high-contrast color palette (Red for fraud, standard colors for baseline data).

- **Geospatial Analysis:** A map visual mapping `Location` against the `Total Fraud Amount`.
- **Merchant Vulnerability:** A Top 10 Clustered Bar Chart highlighting the `MerchantID`s with the highest count of fraudulent transactions.
- **Behavioral Analysis:** Donut charts breaking down fraud by `TransactionType` to see if bad actors are exploiting the "Refund" system or sticking to standard "Purchases."



5. Key Findings & Business Impact

(Note: As this is a portfolio project based on a data snippet, these are the types of insights this specific data model enables you to highlight).

1. **Geographic Hotspots:** The dashboard quickly isolates regions with abnormal transaction behavior. For example, if "New York" shows a disproportionately high fraud rate compared to its total transaction volume, the bank can temporarily tighten regional security parameters.
2. **Merchant Compromise:** By visualizing `MerchantID` vs. `Fraud Volume`, the risk team can identify specific vendors whose payment gateways may have been compromised, allowing the bank to flag future transactions from those merchants for secondary review.
3. **Transaction Anomalies:** The analysis of `TransactionType` reveals whether fraudsters are utilizing stolen cards for direct high-value *Purchases*, or if they are executing complex *Refund* fraud schemes to move money back to secondary accounts.
4. **Card Testing vs. High-Value Theft:** By plotting the `Amount` of fraudulent transactions on a scatter plot, the bank can identify clusters of "micro-transactions" (under $2.00), which heavily indicates automated "card testing" by fraudsters checking if stolen card numbers are active before making larger purchases.

6. Conclusion

The Credit Card Fraud Detection Dashboard successfully bridges the gap between raw data logs and actionable risk intelligence. By automating the calculation of exposure metrics and visualizing the geographic and vendor-based distribution of fraud, the risk management team can drastically reduce investigation times and deploy targeted security measures faster.

7. Future Scope

- **Time-of-Day Analysis:** Create a "Heatmap" matrix in Power BI using the parsed `TransactionDate` time values to identify the most dangerous hours of the week for fraudulent activity.
- **Machine Learning Integration:** Connect the dashboard to a Python script (using Scikit-Learn) to not just visualize the existing `IsFraud` flag, but to actively predict the fraud probability of new, incoming transactions based on historical patterns.